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A Large Language Model-based Framework to Retrieve Life Cycle Inventory and Environmental Impact Data from Scientific Literature

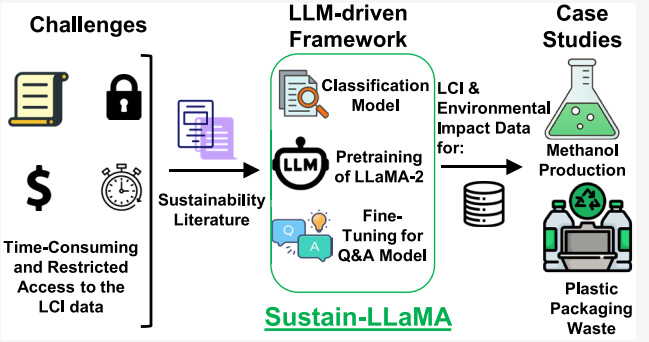
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ABSTRACT: Life cycle assessment (LCA) quantifies environmental impacts from raw material extraction to end-of-life (EoL) treatment, yet its accuracy depends on reliable life cycle inventory (LCI) data. However, obtaining such data is time-consuming and requires an extensive literature review or access to databases that are often behind paywalls that hinder transparent research. This study introduces a systematic framework leveraging a retrained large language model (LLM) to assist LCA practitioners in retrieving LCI data and insightful information about their environmental impact. The framework follows a three-stage process: (i) a fine-tuned classification model identifies relevant documents, (ii) the LLaMA-2-7B model is pretrained on selected texts to inject domain knowledge into its database, and (iii) a fine-tuned Q&A model extracts LCI and environmental impact data from the scientific literature. The resulting LLM is termed as "Sustain-LLaMA". We implement this framework in two cases: methanol production and plastic packaging EoL treatment. After retraining, the classification models achieve high accuracies (0.850 for methanol, 0.952 for plastic packaging) for unseen data, which means effectively distinguishing relevant studies. The Q&A models with Retrieval Augmentated Generation (RAG) yield F1 scores of 0.823 for methanol and 0.855 for plastic studies. The Q&A models' performances are validated against the version of LLaMA-2-7B without retraining, ChatGPT-4o, and the USLCI database, demonstrating comparable or superior accuracy and efficiency. This framework enhances scalability and precision by automating LCI data retrieval, offering a promising tool for guiding the chemical and plastic industries toward sustainability.



KEYWORDS: *life cycle inventory data, environmental impact data, literature mining, large language model, methanol synthesis, plastic packaging EoL treatment*

INTRODUCTION

Evaluating the environmental impacts of various emerging technologies is crucial for identifying optimal alternatives at various life cycle stages to achieve net-zero emissions goals and develop investment strategies for stakeholders.[1] Life cycle assessment (LCA) is a comprehensive tool to estimate environmental impacts, such as greenhouse gas (GHG) emissions, of a product's life cycle. LCA requires life cycle inventory (LCI) data from raw material extraction to end-of-life (EoL) waste treatment. Such data is commonly published in LCA studies in the form of research articles and scientific reports. The amount of such relevant literature is vast, and retrieving practical hidden knowledge requires domain expertise and time, hindering the scalability and automation of LCAs.

Some companies and governmental organizations have attempted to address this gap by storing LCI data in structured databases. The Ecoinvent database[2] is the most comprehensive source, covering a wide range of chemical processes; however, most of its data is aggregated from multiple resources and is accessible only through a paid subscription. Moreover, it restricts the sharing of raw, original LCI data for individual studies, limiting open availability. In contrast, open-source LCI databases such as the U.S. Life Cycle Inventory (USLCI),[3] the European Reference Life Cycle Database (ELCD),[4] and the World Food LCA Database (WFLDB)[5] are freely accessible and include data for diverse industrial processes and agricultural activities. These resources are typically updated by human experts using scientific reports and literature, a process that is time-consuming, costly, and inherently subjective. Consequently, their coverage of the literature is incomplete, leading to gaps in LCI data sets, particularly for regional processes, emerging technologies, and new materials.

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In this study, we showcase the capability of large language models (LLMs) in mining and curating LCA information contained in the long sequential texts of the literature and providing numerical and qualitative insights, a task that might otherwise be time-intensive and tedious for humans. Some available versions of powerful LLMs are GPTs (Generative Pretrained Transformer),[6−8] and LLaMA (Large Language Model Meta AI) versions[9−12] that have been recently released by OpenAI and Meta, respectively. Researchers have developed domain-specific or customized LLM versions in various fields, including material science,[13] biology,[14] geography,[15] chemistry,[16] and life cycle and environmental science.[17] Some more specific applications include Recycle-BERT for plastic recycling,[18] CCU-LLaMA for the carbon capture and utilization literature mining,[19] and for language models to support the hydrogen production.[20,21]

Recent studies have demonstrated the potential of open-source LLMs, such as ChatGPT, GPT-3.5, and GPT-3.5-turbo, to retrieve environmental impact data associated with product life cycles, showing promising results despite certain challenges.[22−24] For example, Deng et al.[26] proposed an Automated Product Carbon Footprint (AutoPCF) framework that uses open-source LLMs as a retrieval engine to identify relevant processes and align them with corresponding emission factors, finding GPT-3.5 to perform comparably well against expert benchmark data. However, AutoPCF's performance was constrained by limited data coverage within open-source LLMs. To address such limitations, researchers have explored the integration of retrieval-augmented generation (RAG) systems as external knowledge bases, which have shown promise in estimating the carbon footprint of products such as aluminum, lithium batteries, photovoltaics, and new energy vehicles.[26,25] Additional efforts include web-based software for exploring biobased construction material alternatives,[27] as well as approaches that optimize low-carbon architectural design by combining LCA with LLMs.[28] Chen et al.[29] further advanced this direction by integrating COMPAS (an open-source computational framework) with GPT-4, enabling automated LCA studies with minimal human intervention and enhanced scalability for architectural projects. Similarly, Chen et al.[30] developed a customized framework that leverages LLaMAIndex to organize relevant literature and applies LLMs (e.g., GPT-3.5-turbo) to retrieve insights on machine learning applications for addressing challenges in LCA studies.

The above-mentioned studies rely on open-source LLMs without employing advanced retraining methods such as pretraining or fine-tuning with specialized domain-specific texts. As a result, the performance of these models is limited, and the accuracy of their outputs is often questionable. The answers provided by these models may include incorrect or hallucinated information and typically require expert review. To date, no study has clearly demonstrated how retrained LLMs can be effectively used to extract environmental impact and Life Cycle Inventory (LCI) data, particularly for products in the chemicals and plastics industry (CPI). This highlights a clear gap in the research. To address this, our work develops a customized LLM-based framework specifically tailored for CPI products.

To enhance the effectiveness of pretraining and fine-tuning, we developed two separate LLMs: one is dedicated to chemicals, for instance, methanol, and the other to plastic products, in particular, plastic packaging. This division is motivated by the unique considerations associated with each category. For example, circularity and waste management play a critical role in efforts toward net-zero emissions in the plastics sector. Initially, maintaining separate models ensures greater specificity and relevance in data extraction and analysis. However, our long-term objective is to integrate these models into a unified framework, enabling a more comprehensive and holistic approach for retrieving life cycle inventory and environmental impact data across the entire chemicals and plastics industries. The chemical case study focuses on retrieving LCI data for methanol, a widely used alcohol in the chemical industry, and an accessible LCI database for easy comparison. Methanol is widely used as a fuel, blended with diesel, and chemical feedstock to support the production of formaldehyde, acetic acid, and MTBE, making it essential in industrial processes. On a large scale, methanol is primarily synthesized from biomass, coal, natural gas, and crude oil.[31] Due to high demand, the extraction and production of these raw materials contribute to 28% of the total carbon emissions from the chemical industry, amounting to 3 billion tons of CO_2 per year for methanol synthesis.[32] The plastics case study focuses on waste from plastic packaging products, whose production is rapidly growing due to their high demand. This sector is one of the major contributors to GHG emissions, which is approximately 3% of total GHG, and plastic waste generation within the plastics economy.[33] The UN Environment Program cautions that by 2040, emissions related to plastics could account for 19% of the global carbon budget.[34] Nearly 40% of total waste belongs to the plastic packaging category due to its short lifespan.[35] While some plastic packaging materials, such as PET bottles, are easily recyclable with conventional mechanical recycling technologies, plastics, such as multilayer packaging films and microplastics, require emerging and more advanced technologies for their recycling and management.[36,37]

We develop a systematic LLM-based framework, namely, "Sustain-LLaMA," which employs retraining techniques such as pretraining and fine-tuning to customize the open-source "LLaMA-2-7B" model for extracting environmental impact and LCI data for chemicals and plastic products from the respective scientific literature. This framework consists of three stages: The first stage involves fine-tuning an LLM to classify relevant articles based on their abstract text, where "relevant" indicates that the text contains information about the product's environmental impact and LCA studies. The second stage utilizes the text from all relevant articles and applies a pretraining technique to inject domain knowledge into the LLM database. Finally, the custom-pretrained LLM is further fine-tuned for a question-and-answer task to develop a user-interactive model, such as a chatbot. This chatbot can be used as an LCA assistant, responding to both qualitative and quantitative inquiries regarding LCA outcomes and LCI data for chemicals and plastics. All three stages are systematically arranged and implemented for two product cases, methanol and plastic packaging, utilizing respective text corpora (scientific literature). We named the respective customized LLMs: "Sustain-LLaMA-Methanol" for methanol and "Sustain-LLaMA-Plastic" for plastic packaging, designed to retrieve environmental impact and LCI data. The novelties and theoretical or application contributions of our proposed LLM-based framework, Sustain-LLaMA, are as follows:

1. A systematic arrangement in the sequence of fine-tuned classification, pretraining, and fine-tuned Q&A stages ensures that the resulting customized LLM is less prone to hallucination and is highly accurate and precise. The

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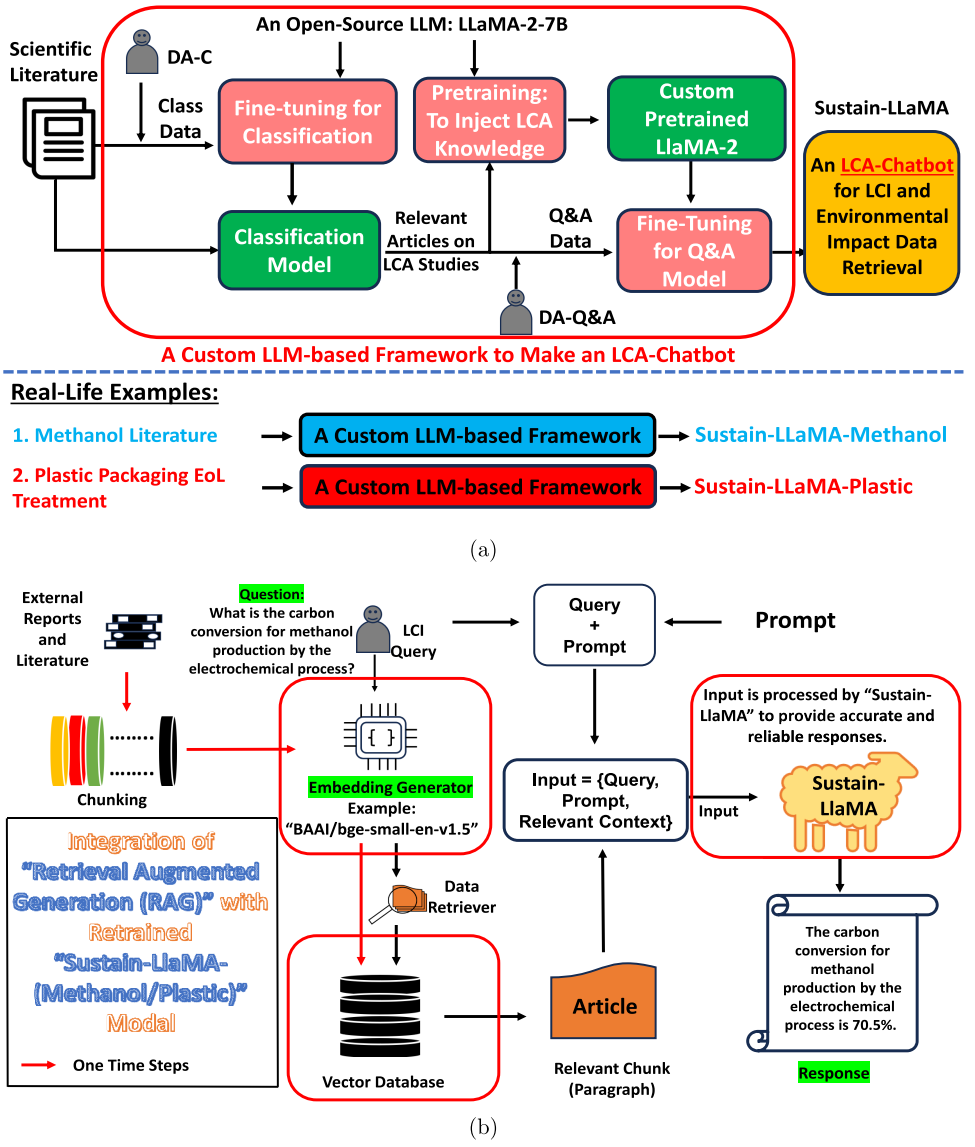


Figure 1. These two figures illustrate (a) a structured retraining workflow to develop Sustain-LLaMA, fine-tuning for classification, pretraining for domain knowledge injection, and fine-tuning for Q&A applied to LLaMA-2-7B using annotated LCA and environmental data sets from two case studies: methanol and plastic packaging end-of-life treatment. (b) A schematic of the RAG system with an LLM: uploaded documents (PDF/TXT) are chunked, vectorized via an embedding generator, and stored in a vector database. An input query is embedded, and relevant chunks are retrieved by a data retriever. The query and retrieved content are then processed by Sustain-LLaMA to generate a source-grounded response.

description is provided in the Subsection of "A Systematic LLM-based Framework: "Sustain-LLaMA".

2. Based on the proposed framework, the resulting LLM is fully capable of answering both qualitative and quantitative queries from users, as discussed in the Section of "Discussion".
3. It can retrieve a range (uncertainty) for the numerical value estimation of environmental impact and LCI data associated with a product's life cycle.
4. The proposed framework is tested on two cases, including methanol and plastic packaging, and it demonstrates its flexibility for other chemicals and plastics as well.

Overall, a thorough use of our customized LLMs provides a broader perspective on the product's environmental impact throughout its life cycle. We provide open access to the Python code on the Mendeley platform,[38] and the resulting customized LLMs on Hugging Face will support other researchers in exploring different products or scaling up these models by retraining them.

The rest of this research article is structured as follows: Section "Data Extraction and Methodology" covers document extraction and summarization of retraining stages, formulated along with discussions on the annotation procedure; the Section "Retraining Results" discusses the retraining results, and the Section "Discussion" shows the performance of both customized LLMs.

DATA EXTRACTION AND METHODOLOGY

Data Extraction. The initial phase of this work involved textual data extraction to develop customized LLMs. Literature data (the corpus) was collected from the Elsevier database, one of the largest scholarly resources, using an Application Programming Interface (API) key, an institute token, and a curated set of influential keywords to ensure comprehensive

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coverage of articles related to LCA and sustainability studies. To facilitate this, two Python scripts were prepared: the first extracted abstracts, DOIs, URLs, publication dates, journal names, country-wise affiliations, and supplementary keywords, while the second retrieved full texts using XML parsing based on the extracted DOIs. The overall data extraction workflow is depicted in Figure S2 in the Supporting file. An iterative approach was followed on the ScienceDirect platform, where keywords were refined by reviewing abstracts and finalizing the most relevant set. In the Supporting file, the selected keywords and related articles are listed in Tables S1 and S2, while Section S1 presents publication trends, leading journals, dominant countries, and keyword clouds for methanol and plastic packaging end-of-life studies.

A Systematic LLM-Based Framework: "Sustain-LLaMA". This section describes the domain-adaptive training along with data annotation steps for developing a customized LLM-based framework using LLaMA-2-7B, which focuses on environmental impact and LCI data for a product's life cycle from scientific literature.

The LLaMA-2-7B model has seven billion hyperparameters with weights that define the model's overall performance, which was optimized based on the training textual data. The pretraining approach effectively optimizes these weights across the entire LLM architecture using new or additional domain text data. Overall, the prior pretraining of LLM with a domain-specific corpus enhances the performance of the fine-tuned model's downstream tasks. The fine-tuning method only updates the weight values of the last layer of LLM in a way that performs only specific tasks, such as classification, question-and-answer (Q&A), and named entity recognition (NER). Although the open-source LLMs are trained on a huge corpus, they still hallucinate responses if they do not have knowledge of specific area-based queries.[39] To prevent hallucinations by using our developed Sustain-LLaMA based on our proposed LLM-based framework, we incorporated both pretraining and fine-tuning approaches.

A systematic arrangement of fine-tuning and pretraining approaches to process domain-specific literature is represented in Figure 1a, resulting in an LLM that aids in extracting environmental and LCI data. The essential steps are described as follows: Initially, a human LCA expert processed a small sample of extracted scientific abstracts and classified them as relevant or nonrelevant. It involved a thorough reading of abstracts and deciding their classes based on the following definitions: The relevant abstracts (annotated as "0") contain information about LCA and environmental studies of a desired product, and nonrelevant abstracts (annotated as "1") do not. This stage of data screening, completed by an LCA expert, took around 1−2 months to annotate the classes to abstracts for both case studies. This annotated data set is named "Class-data." This data set is fed to the classification fine-tuning stage (i.e., the first red box in Figure 1a) of LLaMA-2-7B to classify all the rest of the extracted abstracts into their respective classes. This step is crucial for accurately screening relevant abstracts, saving significant time compared to manual classification, which is prone to human error and limited literature coverage. All classified relevant abstracts' DOIs are used to extract the full text through a fresh Python script as mentioned in the subsection of "Data Extraction." The resulting data set, named "relevant-data," contains the full text of relevant articles, and LLaMA-2-7B is domain-adaptive pretrained to enrich the original LLM database with specific domain data. The full text of the classified relevant abstracts, referred to as "relevant-data," is further processed by the same LCA expert in what we call the question-and-answer (Q&A) curation stage. This stage involves reading the full text of relevant articles and, based on the expert's understanding of a specific product's LCA, curating a set of Q&A pairs. The quality of the curated data set is ensured by using the full text of the relevant articles as a reference to search for answers to the questions. In addition, we incorporated some standard questions with corresponding answers sourced from authorized databases such as USLCI and literature from reputable publishers, including ACS and RSC. We thoroughly reviewed the curated Q&A pairs and found that they are well-aligned for retrieving environmental impact and life cycle inventory (LCI) data for specific products. Overall, the quality and reproducibility of the curated Q&A data sets depend significantly on the level of LCA expertise and the reference resources, such as the relevant articles, used in this study. This stage of processing took approximately the next one to two months to curate; hundreds of question-and-answer pairs were utilized to fine-tune our custom-pretrained LLM to perform question-and-answer tasks for both case studies. Overall, the procedure to annotate the data set was an iterative feedback loop between the first two authors for both tasks, classification and Q&A. We ensured the accuracy of the annotated data sets used in this work. The final model is encapsulated as a Python script with an input function operating like a chatbot that helps retrieve environmental impact and LCI data for a specific domain or product.

In the first stage, the fine-tuned classification model rapidly retrieves relevant articles by analyzing abstracts. Pretraining prior to final Q&A fine-tuning enhances accuracy, as the Q&A model retrieves domain-relevant text and generates reliable responses. Without pretraining, the LLM is prone to irrelevant or hallucinated outputs due to probabilistic word prediction. Figure S5 (Supporting) illustrates quality differences between pretrained−fine-tuned and nonpretrained models for LCI and environmental impact queries. The framework (Figure 1a) is applied to two domains, resulting in "Sustain-LLaMA-Methanol" for methanol and "Sustain-LLaMA-Plastic" for plastic packaging.

In the first stage, the fine-tuned classification model helps to quickly retrieve relevant articles by analyzing the abstracts' text. The pretraining stage, which occurs just before the final fine-tuning stage for the Q&A model, results in a highly accurate, user-interactive Q&A system. This is because whenever a query is submitted to the fine-tuned LLM Q&A model, it retrieves relevant text based on the current knowledge database and then generates a response. However, without a prior pretraining stage, there is a high likelihood that the model may provide irrelevant text, leading to a hallucinated response, as the LLM operates based on probability calculations to predict the next words based on the previous context.

Integration of Retrieval Augmented Generation (RAG). In this subsection, we describe the functioning of Retrieval-Augmented Generation (RAG) with an LLM, which enables access to up-to-date and confidential documents while providing the source of the most relevant contextual information supporting the model's responses. As illustrated in Figure 1b, the process begins with a set of PDF documents, which are divided into manageable chunks and converted into vector embeddings using a sentence embedding generator. These embeddings are stored in a vector database, such as Pinecone. A query is similarly transformed into an embedding, which is then used to retrieve the most contextually relevant chunks from the database. The

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retrieved information, along with the original query, is passed to the LLM, either a custom-trained model such as Sustain-LLaMA-Methanol or Sustain-LLaMA-Plastic or an open-source model like LLaMA-2-7B, which generates a response grounded in the retrieved sources. This integration builds trust in the responses generated by LLMs to each query. Along with this, we can also utilize prompt engineering to direct the LLM to perform specific tasks or respond to queries.

CUSTOM LITERATURE ADAPTIVE RETRAINING RESULTS

Custom-Domain Adaptive Pretraining LLM. The proposed LLM-based framework, "Sustain-LLaMA," is developed using two training approaches for the LLM. The first is the pretraining approach, which allows the model to assimilate domain-specific literature by updating a small portion of the total number of weights across the entire architecture of the LLM; we call it domain-adaptive pretraining. The second is the fine-tuning approach, which adapts the LLM to specific tasks such as classification or question answering by updating only the weights of the final layer based on an annotated data set. The initial stage of the proposed framework involves building a classification model using the fine-tuning approach to screen relevant articles. These articles are then used in the next stage to incorporate domain knowledge into a pretrained (open-source) LLM, resulting in a custom domain-adapted LLM. This adapted model is further fine-tuned for the question-answering task using a question and answer data set. The complete schematic diagram is shown in Figure 1a. The following subsections will present the results of each stage involved in adapting the domain literature to the existing LLaMA-2-7B model, as outlined below: The subsection of "Fine-tuning: Classifying Relevant Articles" covers the classification of relevant articles. The subsection of "Domain Adaptive Pretraining: Injecting Domain Knowledge" discusses the pretraining stage, where domain knowledge is incorporated into the core database, and the subsection of "Fine-tuning: Question and Answer" presents the results of fine-tuning the custom domain-adapted version of LLaMA-2-7B.

Fine-tuning: Classifying Relevant Articles. The custom-annotated classification data set includes two categories of abstracts. It is divided into three subsets: 70% for training, 15% for testing, and 15% for validation. Only the training and validation data sets are used during the fine-tuning stage, while the test data set remains an unseen portion of the annotated data. Binary cross-entropy loss was computed at each step for training and validation, and learning curves were plotted for both case studies.

For the methanol case study, Figure S10a presents the learning curve over three epochs (equivalent to 230 steps), while Figure S10b shows the learning curve for the plastic packaging case, also over three epochs (equivalent to 185 steps). These learning curves show a declining trend for both training and validation, and early stopping was implemented to avoid overfitting. To quantify the overall performance of the fine-tuned models for classification tasks, we utilized four standard metrics: precision, recall, F1-score, and accuracy for predictions on the training and testing data sets. The accuracy values for the test data are 0.850 for the methanol case and 0.952 for the plastic packaging case. These accuracy metric values heavily depend on the accuracy of the annotated data sets, which we ensured. The remaining metric values are provided in Supporting File Section S2, along with the binary loss values for the training and validation data sets.

Domain Adaptive Pretraining: Injecting Domain Knowledge. After successfully classifying all relevant abstracts, we downloaded the full texts of the corresponding articles to inject into the LLaMA-2-7B model's core database, thereby enriching the original database with additional domain knowledge. To overcome these constraints and enable efficient domain-specific adaptation using custom literature, we employed the LoRA (Low-Rank Adaptation) technique, which inserts trainable low-rank matrices into the LLaMA-2-7B model's weight structure. This approach allows fine-tuning a small number of parameters while keeping the base model frozen, effectively enriching the model's knowledge without retraining all weights.[40,41]

Two custom-pretrained LLMs were developed for the two case studies using relevant articles. Retraining was performed on an 82 GB Nvidia GPU at Arizona State University's Sol supercomputer,[42] requiring ten days per case. Learning curves for methanol (Figure S10c) and plastic packaging (Figure S10d) show consistent loss reduction, reflecting improved model understanding of LCA and environmental studies. Final training and validation loss values are provided in Table S5 (Supporting Section S2).

Fine-Tuning: Question and Answer. Using relevant articles, two data sets of questions and their corresponding answers are synthesized by an LCA expert. The data sets consist of hundreds of Q&A pairs, covering both qualitative and quantitative knowledge related to LCI and environmental impact studies. The domain-respecting custom-pretrained LLMs are further fine-tuned using synthesized question-and-answer pair data sets. The respective annotated data set is split into three data types as follows: 70% training, 15% validation, and the remaining 15% as test data.

For example, in the methanol case study, 800 Q&A pairs were used to fine-tune a custom-pretrained LLM over three epochs (equivalent to 540 steps) with a learning rate of 2×10^{-4} and the "AdamW" optimizer.[43] Figure S10e illustrates the trend of training and validation loss values across the 540 steps. Similarly, for the plastic packaging case study, 530 Q&A pairs were curated and used to fine-tune the custom-pretrained LLM. The learning curve, presented in Figure S10f, shows the updating procedure of the LLM's last layer weights across 920 steps. The curves exhibit a consistent downward trend, demonstrating the model's effective learning of contextual meaning from the data set.

The respective fine-tuned models are named as follows: "Sustain-LLaMA-Methanol" for the methanol case, and "Sustain-LLaMA-Plastic" for the plastic packaging case. The performance of these models is evaluated using the BERTScore metric,[44] a semantic evaluation metric that calculates the contextual similarity between the predicted and ground truth responses. It leverages the pretrained language model, Bidirectional Encoder Representations from Transformers (BERT), to generate the embeddings of the responses and then calculates the cosine similarity. Based on similar tokens, it provides three metrics: precision, recall, and F1-score. This metric has some limitations, as follows: it cannot account for numerical fidelity, such as the ability to estimate values within a given tolerance. It also lacks unit conversion capabilities, which are important when environmental impact data (e.g., carbon emissions) are presented in different units, such as 1 g−CO_2−eq versus 0.001 kg−CO_2−eq. Another limitation is its inability to integrate source grounding, meaning it does not distinguish whether a response is supported by citations or not. Subsets of annotated data were fed to both the fine-tuned and base models (i.e., without retraining), and the metrics were calculated based on

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Question: What are the GWP hotspots in the methanol from the natural gas life cycle?

Responses Using:

Sustain-LlaMA-Methanol + RAG: The GWP hotspots in the methanol from the natural gas life cycle are:
1. Natural gas production: **76.6% of the total GWP** is generated during the natural gas production stage.
2. Methanol production: **16.7% of the total GWP** is generated during the methanol production stage.
3. Transportation: **6.6% of the total GWP** is generated during the transportation stage.
4. Use of methanol as a fuel: **0.1% of the total GWP** is generated during the use of methanol as a fuel stage.
Therefore, the GWP hotspots in the methanol from the natural gas life cycle are natural gas production, methanol production, and transportation.

LlaMA-2-7B-BaseModel: The greenhouse gas (GHG) hotspots in the methanol production process from natural gas can be identified as follows:
1. Methane emissions from natural gas production: Methane is the primary component of natural gas, and it is a potent GHG with a global warming potential (GWP) of 28 over a 100-year time horizon. Methane emissions occur throughout the natural gas production process, including during drilling, transportation, and storage. To reduce these emissions, operators can implement methane capture and utilization technologies, such as methane capture and utilization for methanol production.....

ChatGPT-4o: GWP Breakdown by Stage:
Natural Gas Extraction: Methane leaks, fugitive emissions, and energy use (high GWP from CH_4).
Natural Gas Processing and Transportation: Further methane leakage and energy use (moderate GWP).
Steam Methane Reforming (SMR): Direct CO_2 emissions from reforming and high energy consumption (high GWP).
Methanol Synthesis: Moderate energy use and venting of CO_2 (moderate GWP).
Utilities and CO_2 Removal: Indirect emissions from fossil-based energy use and CO_2 venting (moderate GWP).

RAG with LlaMA-2-BaseModel: The GWP hotspots in the methanol from the natural gas life cycle are:
1. Natural Gas Extraction and Transportation: The extraction and transportation of natural gas are the primary sources of GWP emissions in the methanol life cycle. The emissions from these stages are primarily due to the release of methane, a potent greenhouse gas, during the extraction and transportation processes.
2. Coal-to-Methanol (CTM) Process: The CTM process is another significant source of GWP emissions in the methanol life cycle. The process involves the conversion of coal into methanol, which results in the release of GHG emissions, primarily due to the methane content in the coal.
3. Ammonia and Methanol Coproduction (CAM): The coproduction of ammonia and methanol from natural gas results in GWP emissions due to the release of nitrous oxide (N2O), a potent GHG. The N2O emissions are primarily due to the energy required for the process and the use of natural gas as a feedstock.
The above hotspots contribute to the overall GWP of the methanol life cycle, which is estimated to be around 10.6 kg CO2e/kg methanol.

(a)

Question: What are the GWP hotspots in the polyethylene terephthalate life cycle?

Responses Using:

Sustain-LlaMA-Plastic + RAG: The GWP hotspots in the polyethylene terephthalate life cycle are primarily associated with the production of ethylene glycol, which is the primary raw material for PET production. The production of ethylene glycol involves the use of fossil fuels, which contribute to greenhouse gas emissions. Additionally, the production of PET involves the use of energy-intensive processes such as polymerization and extrusion, which also contribute to GWP. The use of PET in packaging and other applications also contributes to GWP due to the energy required for its production and transportation.

LlaMA-2-7B-BaseModel: The life cycle of polyethylene terephthalate (PET) includes several stages, each with its own potential hotspots for greenhouse gas (GHG) emissions and environmental impacts. Here are some of the key GWP hotspots in the PET life cycle: .....

USLCI 2023:

Contribution	Process
⌄ 100.00%	P Polyethylene terephthalate, PET, virgin resin, at plant, kg - RNA
> 74.95%	P Purified terephthalic acid, PTA, at plant, kg - RNA
> 16.26%	P Ethylene glycol, materials production, organic compound, at pla.
> 03.86%	P Natural gas, combusted in industrial boiler - RNA
> 03.83%	P Electricity, at Grid, US, 2010 - RNA

ChatGPT-4o: The main GWP hotspots in the life cycle of polyethylene terephthalate (PET) include: **Raw Material Production**: Producing ethylene glycol and terephthalic acid, the key precursors for PET, is energy-intensive and relies heavily on fossil fuels, contributing significantly to CO_2 emissions. **Polymerization Process**: The polymerization of these raw materials into PET is another energy-intensive step, often powered by fossil fuels, making it a major source of emissions. **Transportation**: Transporting raw materials, intermediates, and finished PET products over long distances adds to the carbon footprint due to fuel consumption. **End-of-Life**: PET incineration releases.....

RAG with LlaMa-2: The GWP hotspots in the polyethylene terephthalate (PET) life cycle are:
1. **Production of virgin PET:** The production of virgin PET is the primary source of greenhouse gas emissions (GHG) in the PET life cycle. The process involves the consumption of fossil fuels, such as naphtha, which releases carbon dioxide (CO2) and other GHGs. According to the study, the production of virgin PET accounts for 70% of the total GHG emissions in the PET life cycle.
2. **Recycling of PET:** While recycling PET can reduce the amount of virgin PET produced, the process itself can also result in GHG emissions. The study found that the recycling of PET generates 1.4 kg CO2-equivalent per kilogram of PET recycled. This is due to the energy required to sort, clean, and re-melt the PET.
3. **Use of PET in packaging:** The use of PET in packaging applications, such as bottles and containers, can also contribute to GHG emissions. The study.....

(b)

Figure 2. A comparison of responses generated by the following models—trained LLaMA-2, Sustain-LLaMA-Methanol (combined with RAG), ChatGPT-4o, base LLaMA-2, and base LLaMA-2 with RAG, is presented for hotspot analysis-related queries for (a) the methanol case and (b) the

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Figure 2. continued

plastic packaging end-of-life treatment case. The response from our developed model, Sustain-LLaMA-Methanol, is supported by,[47] while the response from Sustain-LLaMA-Plastic is supported by,[48] which is the most relevant article to the queries.

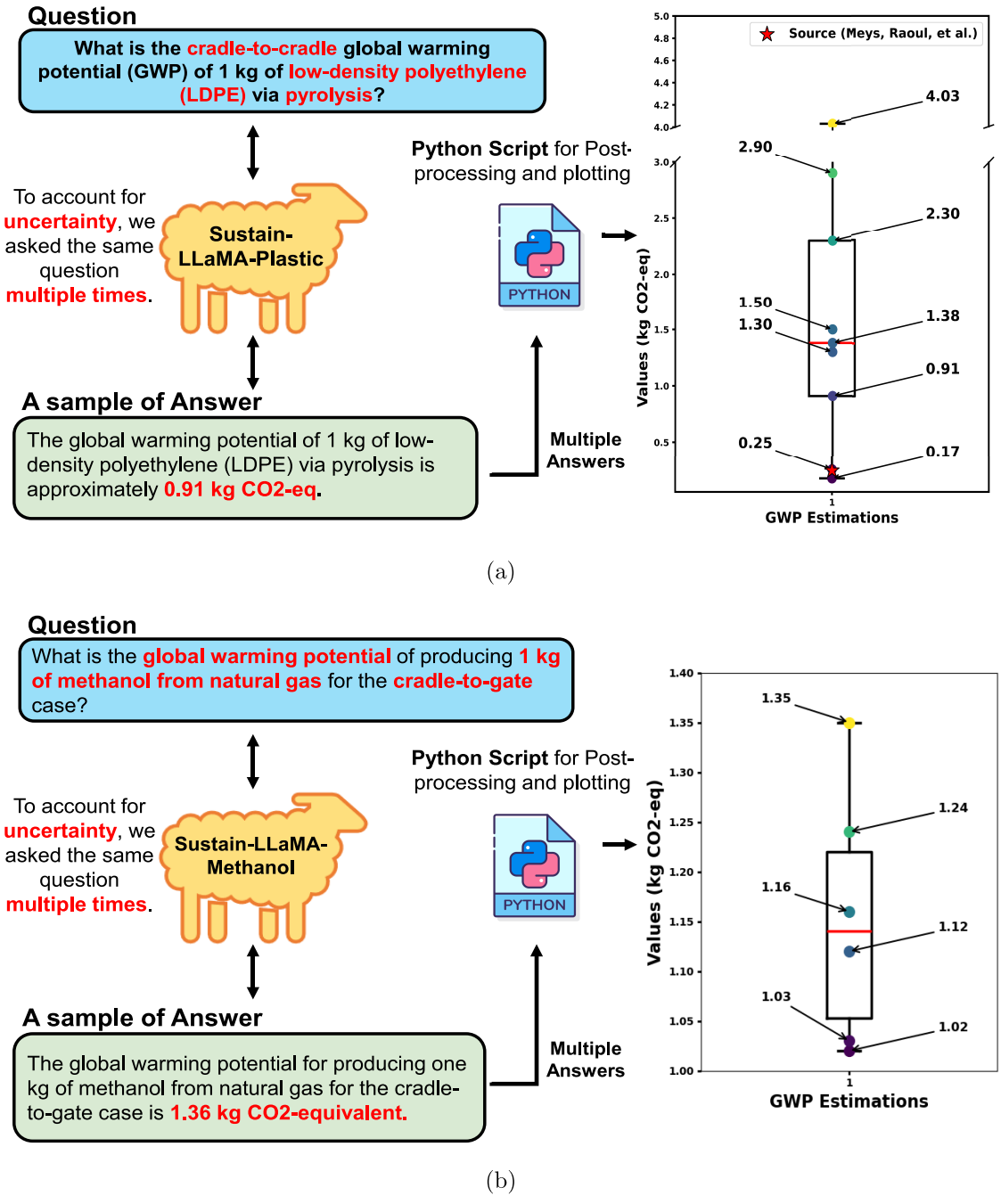


Figure 3. Diagrams depict how environmental impact data, including uncertainty ranges, which stand for aleatory uncertainty associated, are retrieved for global warming potential (GWP) in two scenarios: (a) methanol production assessed from cradle to gate, using natural gas as the feedstock and validated with the USLCI database and relevant article[49] (b) LDPE assessed from cradle to cradle, incorporating pyrolysis as the end-of-life method and verified through theoretical calculations from.[50] Both scenarios present the data retrieval steps and corresponding error box plots to represent uncertainty.

the ground truth and predicted responses. For the testing data set, the F1-score for the methanol case was 0.823 for the fine-tuned model with RAG and 0.798 for the base model with RAG. For the plastic packaging case, the fine-tuned model with RAG achieved an F1-score of 0.855, while the base model scored 0.821, which is approximately 3−4% lower than the retrained LLM's performance. The values for the remaining metrics are presented in the Supporting file, Section S2. The details of the hyperparameters, data size utilized, and computing time for each stage of retraining are also tabulated in Tables S6 and S7 in

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Section S3 of the supporting file. These models serve as LCA assistants, retrieving LCI and impact data for methanol life cycles and plastic packaging end-of-life treatments via queries.

DISCUSSION

We evaluated the retraining performance of fine-tuned and pretrained LLaMA-2−7B models for classification, domain-specific pretraining, and Q&A across two case studies. The resulting models, Sustain-LLaMA-Methanol and Sustain-LLaMA-Plastic, were tested using LCA expert queries. Performance was assessed through quantitative queries with expected numerical outputs and qualitative queries requiring concise, specialized insights. Results were benchmarked against open-source databases (e.g., USLCI) and literature, comparing Sustain-LLaMA (with RAG), the base LLaMA-2−7B with RAG, and ChatGPT-4o. The resulting quantitative and qualitative responses were verified by an LCA expert, who also curated a set of queries for both case studies (Section Performance of "Performance of Sustain-LLaMA-Methanol" for methanol and Section of "Performance of Sustain-LLaMA-Plastic" for plastic packaging). To ensure reliability, the retrained models were first validated against the expert's knowledge, relevant literature, and the external LCI database (USLCI). Their outputs were then systematically compared with results from ChatGPT-4o and the baseline LLaMA model without retraining.

Performance of Sustain-LlaMA-Methanol. In this subsection, we summarize the performance of the domain-retrained LLM in retrieving both quantitative LCI data and qualitative responses for methanol's life cycle. We evaluated the model using questions designed for qualitative analysis, integrating RAG systems as discussed in Section "Integration of RAG." Figure 2a compares responses from "Sustain-LLaMA-Methanol" with RAG, "LLaMA-2-BaseModel," "LLaMA-2-BaseModel" with RAG, and "ChatGPT-4o" for a hotspot analysis query on methanol synthesis from natural gas. Our fine-tuned model with RAG provided source-backed, detailed numerical and descriptive responses, unlike the other models. Additional expert-generated technical questions showed similar trends (see Section S4, Figures S10−S12). To maintain accessibility, queries were kept simple, yet responses remained accurate, suggesting that more detailed queries could yield even richer insights.

After succeeding in the comparison of responses, we analyzed the global warming potential (GWP), which is an essential metric for measuring the environmental impact associated with any product's life cycle. The reported GWP values have a wide range for methanol's life cycle due to the following major reasons: different raw materials used (e.g., biomass, coal, and natural gas), technologies/processes, assumptions, and system boundaries (e.g., gate-to-gate, cradle-to-gate, and cradle-to-cradle). To demonstrate the effectiveness of our customized LLM in retrieving the GWP value range, we formulated a query based on the following parameters: natural gas as the raw material, a cradle-to-gate system boundary, and the production of one kilogram of methanol. The GWP value varies even with such constraints because different LCI data come from different resources, contributing to an uncertainty in the retrieved GWP value. When using a trained LLM, the hyperparameter temperature controls the degree of variability in its predictions (or responses). A higher temperature introduces more diverse and uncertain responses, while a lower temperature produces more deterministic outputs. We assume that such variations, which reflect uncertainty across different technologies, can be relevant in synthesis and treatment processes during life cycle assessment (LCA). Even when the same raw material, system boundary, or query is used, the stochastic nature of the LLM allows it to capture this diversity. This uncertainty requires further investigation, drawing on the provided sources and the responses generated by the trained model using RAG. To reflect the expected uncertainty in our study, we set the temperature to 0.7, based on our testing experience with the model's responses. Using our developed LLM, we submitted the same formulated query multiple times, and the resulting responses were stored and analyzed to produce the box plot shown in Figure 3a. If we keep this hyperparameter value at zero, it acts like a single random seed. Figure 3a contains the formulated query and a schematic flow for a single run. The box plot represents the distribution of GWP values; a range of 1.05 to 1.35 kg $CO_{2\text{-eq}}$ is observed, and this range is validated against a standard GWP value of 1.16 kg $CO_{2\text{-eq}}$ from the USLCI database,[3] as presented by a red star mark, using the same constraints mentioned in our simplified query. This range of values is also validated through using the RAG system with sustain-llama-methanol, and found relevant citations for them as mentioned in the caption of Figure 3. Overall, the above-stated argument justifies the model's performance accuracy for one of the essential environmental impact measuring metrics.

After validating Sustain-LLaMA-Methanol through qualitative and quantitative analysis, we extracted basic LCI data under a gate-to-gate boundary with syngas as the methanol feedstock. Multiple queries were processed, and results with associated uncertainties (mean ± standard deviation) are summarized in Table 1. Retrieved values were consistent with published data.

Table 1. Extracted Life Cycle Inventory Data, with Uncertainty Considerations for the Methanol Synthesis Process, Use Syngas Composition as Input Variables within A Gate-to-Gate System Boundary and Are Compared to Published Work[45,46]

Input Variable Name	Outcome using "Sustain-LLaMA-Methanol" (kg unit)	Standard LCI Database & Stoichiometric Calculations
H_2 required	1.31 ± 0.120	1.25
CO_2	1.35 ± 0.621	1.375
CO	0.679 ± 0.435	0.875
H_2O	1.30 ± 0.75	0.560

For example, hydrogen consumption for CO_2 hydrogenation to methanol was reported as 1.25 kg/kg MeOH, while our model yielded 1.31 ± 0.12 kg/kg MeOH. Similar consistency was observed forCO_2, CO, and H_2O inputs, confirming the model's ability to reliably retrieve LCI data for this system boundary.[45,46]

Performance of Sustain-LLaMA-Plastic. Similarly, the Sustain-LLaMA-Plastic model is evaluated against BaseModel, ChatGPT-4o, and the USLCI database for hotspot analysis of PET plastic life cycle (Figure 2b). Our model uniquely provides accurate numerical entities, verified using the USLCI database, while other models lack such data. Additional examples in Section S3 (Figures S12−S16) show that Sustain-LLaMA-Plastic generates more concise and informative responses than the base model and ChatGPT-4o. Overall, the model effectively addresses inquiries on GWP hotspots in the plastic supply chain, polymer production, and EoL processing, as referred to in Section S3 of the supporting file.

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Next, we evaluate the performance of our customized LLM in retrieving the GWP value range for plastic packaging. We constructed a query based on the following criteria: a cradle-to-cradle system boundary, low-density polyethylene (LDPE), and pyrolysis as an end-of-life treatment technology. Although the criteria were held constant, the calculated GWP values still varied due to differences among life cycle inventory (LCI) data sources corresponding to chemical recycling via pyrolysis. In other words, greenhouse gas emissions from pyrolysis are influenced by several factors, including the purity and contamination level of the feedstock, the operational conditions of the pyrolysis reactor, and the energy mix used to supply heat and electricity, among others. We repeated the same query multiple times to evaluate how well our LLM captures this variability. The outputs were collected and used to construct the box plot in Figure 3b, which also displays the query text and an example response. This plot shows the distribution of GWP values retrieved, which we verified by comparing them with a reference value from literature[50] under the same conditions. Figure 3b shows a broader range of GWP values, from 4.03 to 0.17 kg $CO_{2\text{-eq}}$, under a cradle-to-cradle boundary condition. This includes pyrolysis as an end-of-life treatment technique, which requires an energy-intensive setup. Variations in the type of setup may cause different GWP values at this stage of treatment. We validated this range of GWP values using a reported value of 0.24 kg $CO_{2\text{-eq}}$ by Meys et al.[50]

We evaluated our customized LLM for retrieving GWP values of plastic packaging under a cradle-to-cradle boundary, focusing on LDPE with pyrolysis as the EoL treatment. Despite fixed criteria, retrieved values varied due to differences in LCI data and factors such as feedstock purity, contamination, reactor conditions, and energy mix. We repeated the query to test consistency, and the outputs were compiled into the box plot in Figure 3b, which also shows the query and an example response. The GWP values ranged from 0.17 to 4.03 kg $CO_{2\text{-eq}}$, reflecting the energy-intensive nature of pyrolysis. This range was validated against a reference value of 0.24 kg $CO_{2\text{-eq}}$ reported by Meys et al.[50]

In summary, we proposed an emerging large language model-based framework to access life cycle inventory and environmental impact category data, such as global warming potential, for the chemicals and plastic products present in scientific literature. We demonstrated it by application to two case studies: methanol and plastic packaging end-of-life treatment. We developed two respective question-and-answer models for these two cases, "Sustain-LLaMA-Methanol" and "Sustain-LLaMA-Plastic," for methanol and plastic packaging, respectively. The models are capable of responding to quantitative and qualitative inquiries related to life cycle analysis. The models' performance is evaluated using multiple schemas for quantitative and qualitative analysis of responses, including ChatGPT-4o, the base LLaMA-2−7B (which is not retrained), and the open-access LCI database, USLCI. We found that our models perform well compared to benchmark models, demonstrating both accuracy and quality of responses. The F1-score metric is calculated to assess the contextual similarity between the predicted and generated responses. We found that "Sustain-LLaMA-Methanol" achieved values close to 0.823, while "Sustain-LLaMA-Plastic" achieved values ranging between 0.855 and 0.872, irrespective of the training and testing data sets. The developed LLMs are also integrated with a retrieval-augmented generation (RAG) system, which enables access to the sources of responses and external documents that are more up-to-date and confidential. The current evaluation criteria, such as quantitative and qualitative analysis, can be further extended, for example, by enabling back-tracing of response sources for additional verification. For the same reason, all Python code and models developed in this study are openly available, enabling researchers to build upon and adapt the framework for other chemicals. For example, the methanol-based model can be extended to other alcohol-class chemicals, such as ethanol and butanol, due to shared scientific terminology, supporting the development of an alcohol-focused LLM. This approach can similarly be applied to other classes, including the concrete, glass, and steel industries; plastics and materials; and the energy sector. The proposed framework was validated using two case studies and can be readily extended to other chemical systems by the community. To develop a more robust and optimized version of these LLMs, we will extend this work to enhance overall performance and create a new generation of LLMs specialized for various tasks related to plastics and chemicals. It is also possible to conduct a sensitivity analysis using different training approaches and to evaluate the impact of data size on model performance before releasing an updated version of the LLM. Additionally, we propose developing a hallucination benchmark data set through an in-depth analysis of LLM-generated responses using a document verification technique within the RAG system, followed by an evaluation of the LLMs' hallucination tendencies using the same curated database. Overall, this work aims to help LCA practitioners gain easier access to environmental impact data and life cycle inventory data for a product's life cycle. These customized LLMs can be encapsulated into a user-friendly, open-access chatbot tool integrated with retrieval-augmented generation (RAG), especially knowledge graph-based RAG (Graph-RAG), which helps to retrieve more accurate and concise responses to the queries with its source. This tool would enable users to upload confidential or newly generated documents and reports, allowing them to efficiently retrieve information and extract relevant insights in minimal time. This approach presents a promising avenue for enhancing the performance of future use cases.

■ ASSOCIATED CONTENT

Data Availability Statement

The Python code synthesized in this proposed study is now open access on the Mendeley platform,[38] along with all required information and the text data sets utilized. We have uploaded all trained versions of Sustain-LLaMA for methanol and plastic packaging EoL treatment case studies on the authors' https://huggingface.co/avankumar Hugging Face platform.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.5c05955.

Section S1, auxiliary details about extracted data, such as subsection S1.1 for Finalized keywords statistics and subsection S1.2, Statistical overview of research trends and contributions; Section S2, as the performance metrics evaluation; Tables S3 to S5 represent the tabulated metric values for all three tasks; At last, Section S3, showcases the performance of the resulting Q&A LLM versions regarding methanol and plastic packaging end-of-life treatment case studies (PDF)

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Notes

The authors declare no competing financial interest.

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